

Numerical methods

Problems for the practices

Week 2 – Errors, error propagation

2.1. Find the absolute and relative error bounds for the following approximations!

(a) $\sqrt{3} \approx 1.73$ (using 2 significant digits);

(b) $\sqrt{3} \cdot \sqrt{3} \approx 1.73 \cdot 1.73$;

(c) $\pi \approx 3.14$;

(d) $\frac{1}{\pi} \approx \frac{1}{3.14}$.

2.2. In six-digit rounding arithmetic calculate the following operations! (Use base 10 with mantissa length $t = 6$.)

$$\sqrt{2021} - \sqrt{2020} \quad \text{and} \quad \frac{1}{\sqrt{2021} + \sqrt{2020}}$$

Explain which one of the two formulas gives a better estimation!

2.3. Give an algorithm for the solution of the quadratic equation

$$x^2 - 2px - q = 0 \quad (p, q > 0)!$$

What can we say if $p \gg q$? Consider the example

$$x^2 - \sqrt{2}(10^5 - 10^{-5})x - 2 = 0.$$

2.4. The relative error can be defined in two different ways:

$$\frac{\Delta a}{A} \quad \text{and} \quad \frac{\Delta a}{a}.$$

How much is the difference between them?

2.5. Error of functions. Let

$$f(x) = \frac{x}{1 - x^2} \quad (x \neq \pm 1).$$

We calculate the value of $f(x)$ where $x \approx 2$. Find the absolute error bound $\Delta_{f(x)}$ if $\Delta_x = 0.1$! What is the condition number of the function f at this point?